

● Original Contribution

POSTOPERATIVE RADIATION THERAPY FOR INCOMPLETELY RESECTED COLORECTAL CARCINOMA

PAUL E. ALLEE, JOEL E. TEPPER, LEONARD L. GUNDERSON AND JOHN E. MUNZENRIDER

Radiation Medicine Service, Massachusetts General Hospital Cancer Center and Harvard Medical School, Boston, MA

The efficacy of high-dose post-operative radiation therapy was evaluated in 56 patients with pathologically proven or suspected residual disease after surgical resection of colon or rectal carcinoma. Patients had either microscopic or gross residual. They were treated with pelvic or abdominal irradiation to a dose of 4500 cGy followed by boost therapy to as much as 6000 to 7000 cGy if small bowel could be moved from the radiation field. Patients with microscopic residual had a local failure rate of 30% compared to 57% in those with gross residual disease. Five-year disease-free survival was 45% in patients with microscopic versus 10.6% for those with gross residual tumor. There was a trend toward a dose response curve for those with microscopic disease, but none was noted with gross residual. In view of the limited results obtained with current external beam techniques, it is recommended that newer avenues, such as high-dose preoperative therapy combined with intraoperative radiation, be investigated in this poor prognosis group.

Colorectal cancer, Incomplete resection, Post-operative radiotherapy, Microscopic residual, Gross residual.

INTRODUCTION

Adjuvant post-operative radiation therapy for rectal and colonic adenocarcinoma is now frequently used for patients who, despite apparent complete resection, are considered at high risk for local recurrence (1, 6, 9, 14). The usual criteria for offering such therapy include transmural tumor extension and/or regional lymph node metastases. A number of studies have indicated that post-operative irradiation can be effective in reducing the incidence of local recurrence when given in an adjuvant setting (4, 12, 17).

Patients in whom tumor resection is thought to be incomplete represent a more difficult problem for the radiation oncologist. In this situation it is necessary to be able to sterilize a larger bulk of tumor and more clonogenic cells than when radiation is used in a strictly adjuvant setting. However, the role of post-operative irradiation in patients with proven or presumed residual disease has received relatively little attention in the literature (5, 13). This analysis was undertaken to examine the problem of residual disease in more detail, to determine the efficacy of post-operative irradiation in such cases, and to suggest possible strategies for better local control.

METHODS AND MATERIALS

From the Tumor Registry of the Massachusetts General Hospital 63 patients were identified who received radiation therapy between 1974 and 1980 for residual colorectal carcinoma. Patients were deemed acceptable for inclusion on this analysis if (a) there was biopsy proven evidence of disease remaining after resection (pathologic gross residual), (b) the surgeon described gross disease remaining after resection even if a biopsy was not performed (surgically presumed), (c) tumor was dissected sharply off an adjacent unresected structure (operative microscopic residual), or (d) the margins of resection were positive for tumor (pathologic microscopic residual). No patients had evidence of metastatic disease beyond the regional lymph nodes. Surgical resection varied depending on the location of the primary tumor but generally consisted of a partial colectomy for colon cancer, or an abdominoperineal resection or anterior resection for rectal cancer. Thirty-five patients were male and 28 female; the age range was 39-90 with a median of 65 years.

Since 1976, radiation techniques have been standard (8). All patients received megavoltage external beam therapy, usually with 10 MV photons. In general, multiple

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Reprint requests to: Paul E. Allee, M.D., Dept. of Radiation Oncology, Tufts-New England Medical Center, 750 Washington Street, Boston, MA 02111.

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